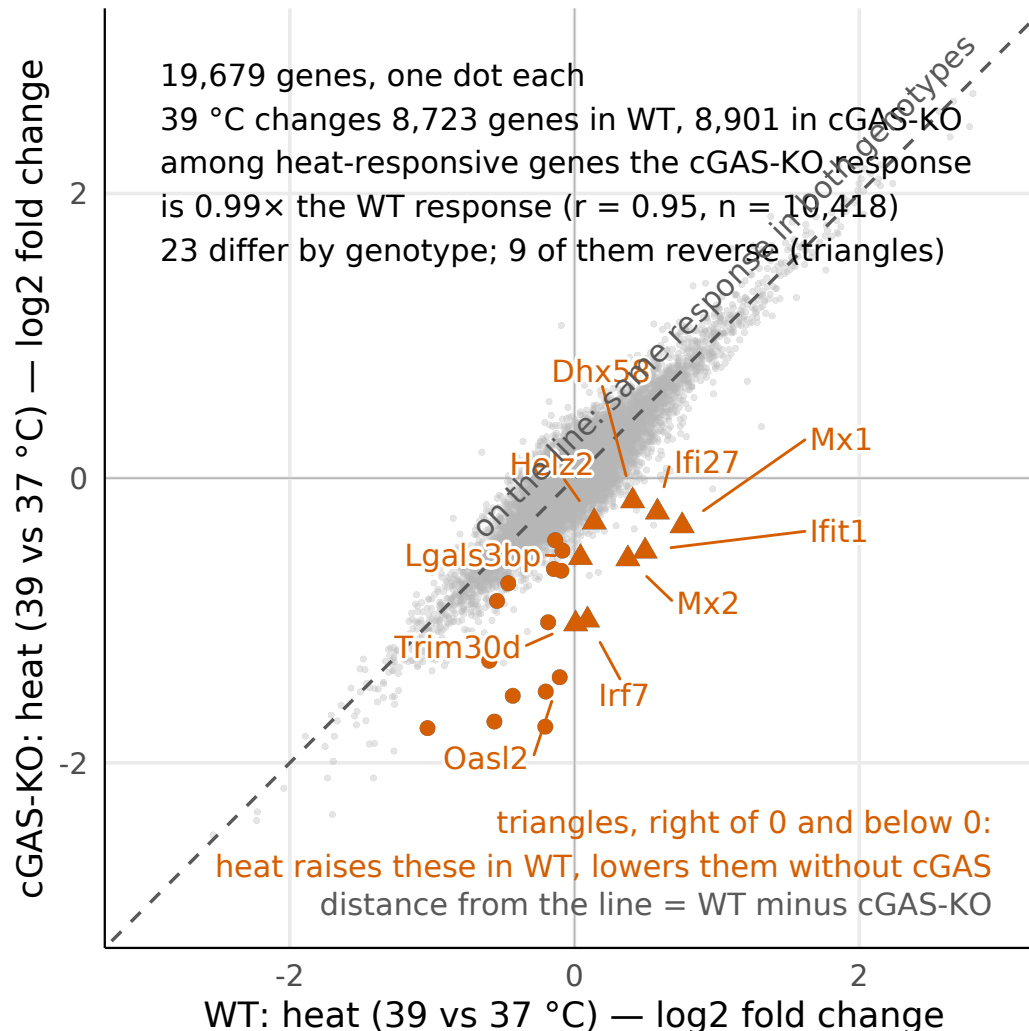


Warming to 39 °C changes the same genes with and without cGAS — apart from one interferon arm

Each dot is one gene. Both axes are its log2 fold change at 39 versus 37 °C, measured in WT (x) and in cGAS-KO (y). The dashed line marks an identical response in the two genotypes.



- no detectable difference between genotypes at $n=5$
- heat response differs between genotypes ($\text{adj.}P < 0.05$)
- ▲ reverses direction: up with heat in WT, down without cGAS

All 23 genes in the arm respond to heat more strongly in WT than in cGAS-KO; 0 do the reverse. Every one of them falls with heat once cGAS is gone (22 of 23 individually significant), and 9 rise with heat in WT instead -- an interferon response that heat sustains only when cGAS is present. The genotype comparison of the heat response is a single-degree-of-freedom test at $n=5$ per group, the least-powered contrast in this design, so 23 genes is a floor on the cGAS-dependent arm rather than its full size. Genes on the shared axis have no detectable cGAS-dependence at $n=5$, which is a weaker statement than independence.